

1 Measure theory

We recap some basics of measure and probability theory following the structure of [3], mathematical aid can be found in [4] or watch [1].

1.1 Basic Definitions

- **σ -algebra:** Given a set X , the σ -algebra $\mathcal{F}$ on X is a collection of subsets of X , $2^X \subseteq \mathcal{F}$ such that

1. $X \in \mathcal{F}$,
2. It is closed under complementation: $A \in \mathcal{F} \Rightarrow X \setminus A \in \mathcal{F}$,
3. It is closed under countable unions: $A_1, A_2, A_3, \dots \in \mathcal{F} \Rightarrow A_1 \cup A_2 \cup A_3 \cup \dots \in \mathcal{F}$,

where 2^X is the power set of X . Apparently $\emptyset \in \mathcal{F}$ and as a consequence also countable intersections belong to $\mathcal{F}$: $A_1, A_2, A_3, \dots \in \mathcal{F} \Rightarrow A_1 \cap A_2 \cap A_3 \cap \dots \in \mathcal{F}$. Apparently $\{\emptyset, X\}$ is the smallest and 2^X the largest σ -algebra of Ω .

- The tuple $(X, \mathcal{F})$ is called **measurable space**.
- **Measure.** Let $(X, \mathcal{F})$ be a measurable space. A measure $\mu : \mathcal{F} \rightarrow [0, \infty]$ is a function satisfying
 1. Non-negativity: $\mu(A) \geq 0 \quad \forall A \in \mathcal{F}$
 2. Null empty set: $\mu(\emptyset) = 0$
 3. Countable additivity (σ -additivity). If $\{A_i\}_{i=1}^\infty$ are *disjoint* sets in $\mathcal{F}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

A **probability measure** additionally satisfies

4. $\mu(X) = 1$

- The triple $(X, \mathcal{F}, \mu)$ is called **measure space**.

1.2 Relation between measurable spaces.

- **Measureable function.** Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be two measurable spaces. A function $f : X \rightarrow Y$ is called measurable if $f^{-1}(B) \in \mathcal{F}$, $\forall B \in \mathcal{G}$. So a measureable function respect the σ -algebra structure (pull backs measurable sets in it's codomain to measurable sets in it's domain). One also says $\mathcal{F} - \mathcal{G}$ (or, $\mathcal{F}/\mathcal{G}$) measurable to stress the underlying σ -algebras.
- **σ -subalgebra:** Let $(\Omega, \mathcal{F})$ be a measurable space with σ -algebra $\mathcal{F}$ on Ω . An other σ -algebra $\mathcal{G}$ on Ω is called σ -subalgebra of $\mathcal{F}$ if $\mathcal{G} \subseteq \mathcal{F}$.

¹For $A_1, A_2 \in \mathcal{F} \Rightarrow X \setminus A_1, X \setminus A_2 \in \mathcal{F} \Rightarrow (X \setminus A_1) \cup (X \setminus A_2) \in \mathcal{F} \Rightarrow X \setminus ((X \setminus A_1) \cup (X \setminus A_2)) = A_1 \cap A_2 \in \mathcal{F}$, where the last equality follows from De Morgan's law.

- **Filtration.** Given a measurable space $(X, \mathcal{F})$. A filtration is a family of σ -subalgebras $(\mathcal{F}_t)_{t \geq 0}$ such that $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$, for all $s \leq t$.

In a *stochastic process* this is used to interpret $\mathcal{F}_t$ as the collection of events observable up to time t and $\mathcal{F}_s \subseteq \mathcal{F}_t$ implies that information does not decrease over time (knowledge at time s is also available at time t). A stochastic process (M_t) is called adapted to the filtration $(\mathcal{F}_t)$ if M_t is measurable with respect to $\mathcal{F}_t$.

- **Intersection of σ -algebras is a σ -algebra.** Let $\{\mathcal{F}_i\}_{i \in I}$ be a family of σ -algebras of X (countable or not). Define the intersection as

$$\bigcap_{i \in I} \mathcal{F}_i = \{A \subseteq X \mid A \in \mathcal{F}_i \forall i \in I\}$$

- **σ -algebra generated by set.** The *smallest* σ -algebra generated by a set $\mathcal{C} \subseteq 2^X$ is the intersection of all σ -algebras containing $\mathcal{C}$.

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{F} \mid \mathcal{C} \subseteq \mathcal{F}, \mathcal{F} \text{ is } \sigma\text{-algebra}\}$$

Note the $\mathcal{C}$ is a family of subsets of X . This is extremely useful for constructing σ -algebras for non-finite sets (and constructing measures). For example the power set of $\mathbb{R}$ is a beast (still a σ -algebra) and it is much more convenient to work with the σ -algebra generated by all open sets of $\mathbb{R}$, called the Borel σ -algebra $\mathcal{B}(\mathbb{R})$.

- **σ -algebra generated by a function.** Let X be a set and $(Y, \mathcal{G})$ a measurable space and a function $f : X \rightarrow Y$. Denote the set of preimages by $\mathcal{C} = \{f^{-1}(B) : B \in \mathcal{G}\}$. The σ -algebra generated by f on X is the σ -algebra generated by the set of preimages $\sigma(f) := \sigma(\mathcal{C})$. Apparently it is the smallest σ -algebra on X that makes f measurable.
- **Product of σ -algebras** Let $(X_1, \mathcal{F}_1), (X_2, \mathcal{F}_2)$ be measurable spaces, then the product sigma algebra is defined by

$$\begin{aligned} \mathcal{F}_1 \otimes \mathcal{F}_2 &:= \sigma(\mathcal{F}_1 \times \mathcal{F}_2) \\ \mathcal{F}_1 \times \mathcal{F}_2 &= \{A_1 \times A_2 \mid A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2\}, \end{aligned}$$

That is the σ -algebra generated by the set $\mathcal{F}_1 \times \mathcal{F}_2$ on $X_1 \times X_2 = \{(x_1, x_2) \mid x_1 \in X_1, x_2 \in X_2\}$ ². Thus the product of measurable spaces can be declared as $(X_1 \times X_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$. Because $(\mathcal{F}_1 \otimes \mathcal{F}_2) \otimes \mathcal{F}_3 = \mathcal{F}_1 \otimes (\mathcal{F}_2 \otimes \mathcal{F}_3)$ it is enough to write $\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes \mathcal{F}_3$. Furthermore, for Borel sets things are smooth, $\mathcal{B}(\mathbb{R}^{p+q}) = \mathcal{B}(\mathbb{R}^p) \otimes \mathcal{B}(\mathbb{R}^q)$.

- **Pushforward measure.** Let $(X, \mathcal{F}, \mu)$ be a measure space, $(Y, \mathcal{G})$ a measurable space, and $f : X \rightarrow Y$ a measurable function. The pushforward measure of μ under f , denoted by $f_*\mu$ (or $\mu \circ f^{-1}$) is a measure on $(Y, \mathcal{G})$ defined by $f_*\mu(B) = \mu f^{-1}(B)$, $\forall B \in \mathcal{G}$.³

² $\mathcal{F}_1 \times \mathcal{F}_2$ is not a σ algebra. This can be seen from the counterexample $\mathcal{F}_1 = \mathcal{F}_2 = 2^{\{1,2\}}$

³The pushforward measure is indeed a measure. Countable additivity follows from $B_i \cap B_j = \emptyset \Rightarrow f^{-1}(B_i) \cap f^{-1}(B_j) = \emptyset$, $B_i, B_j \in \mathcal{G}$, that is disjointness carries over to the preimages. To show this assume the contrary, $B_i \cap B_j = \emptyset$ but $f^{-1}(B_i) \cap f^{-1}(B_j) \neq \emptyset$. Thus there exists $\omega \in f^{-1}(B_i)$ but also $\omega \in f^{-1}(B_j)$, therefore $f(\omega) \in B_i$ and $f(\omega) \in B_j$ and therefore $\omega \in B_i \cap B_j \neq \emptyset$, which is a contradiction.

1.3 Probability theory

- A measurable space is commonly written as $(\Omega, \mathcal{F})$, where Ω is also called **sample space** (the set of all outcomes).
- A measure space $(\Omega, \mathcal{F}, p)$ with p a probability measure is called **probability space**. Probability measures are also called **probability distributions**. For a probability measure the complement follows as⁴ $p(\Omega \setminus A) = 1 - p(A)$.
- Given a probability space $(\Omega, \mathcal{F}, p)$, a measurable function X is called **random variable** if $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, **random vector** if $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ and **random element** $X : (\Omega, \mathcal{F}) \rightarrow (\mathcal{X}, \mathcal{G})$, where $(\mathcal{X}, \mathcal{G})$ is a measurable space. So a random variable is a random element with the specification that the codomain is the measurable space of the real line together with the Borel σ -algebra.

Because X is measurable, the pushforward measure $X_*p = p \circ X^{-1} := p_X$ can be canonically assigned to render the codomain a probability space in either case. The pushforward measure is also called **law**.

Note, that capital letters are common in probability theory for functions, which encourages confusions. Furthermore, the canonical existence of a pushforward measure gives rise to further confusion ('probability distribution of a random variable' instead probability distribution induced by a random variable). Furthermore, as all questions can be answered from p_X alone this is often used as an excuse to not mention the underlying probability space $(\Omega, \mathcal{F}, p)$ at all.

Since random variables are functions, one can consider compositions, and algebraic operations on the real line of them.

Example 1. *Coin toss. Define the probability space $(\Omega, \mathcal{F}, p)$ by*

$$\begin{aligned}\Omega &= \{H, T\}, \\ \mathcal{F} &= 2^\Omega = \{\{H\}, \{T\}, \{H, T\}, \emptyset\}, \\ p(A) &= \begin{cases} \frac{1}{2}, & A = \{H\}, \{T\} \\ 1, & A = \{H, T\}, \\ 0, & A = \emptyset, \end{cases}\end{aligned}$$

where $A \in \mathcal{F}$. Define the random variable

$$\begin{aligned}X : \Omega &\rightarrow \mathbb{R} \\ \omega &\mapsto \begin{cases} 1, & \omega = H \\ 0, & \omega = T \end{cases}\end{aligned}$$

We need to show that the function X is a random variable, i.e., it is a measurable function to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Pick a set $B \in \mathcal{B}(\mathbb{R})$. Then $X^{-1}(B) = \emptyset$ if neither 1

⁴ $\Omega = A \cup \Omega \setminus A \Rightarrow p(\Omega) = \mu(A) + p(\Omega \setminus A) \Rightarrow p(\Omega \setminus A) = 1 - \mu(A)$ by property $p(\Omega) = 1$ of a probability measure.

or 2 are in B , $\{H\}$ if $1 \in B$, $\{T\}$ if $0 \in B$, $\{H, T\}$ if both are in B . In either cases $X^{-1}(B) \in \mathcal{F}$. So indeed X is a random variable.

For the pushforward measure note that singletons are in $\mathcal{B}$ and thus countable unions of it and thus (subsets of) the natural numbers. Thus $p_X(\{1\}) = p_X(\{2\}) = 1/2$.

Example 2. Rolling an even number with a die. Define the probability space $(\Omega, \mathcal{F}, p)$ by

$$\begin{aligned}\Omega &= \{1, 2, 3, 4, 5, 6\}, \\ \mathcal{F} &= 2^\Omega \\ p(A) &= \frac{|A|}{|\Omega|} = \frac{|A|}{6}, A \in \mathcal{F}\end{aligned}$$

Define the random variable

$$\begin{aligned}X : \Omega &\rightarrow \mathbb{R} \\ \omega &\mapsto \begin{cases} 1, & \omega \text{ even} \\ 0, & \omega \text{ odd} \end{cases}\end{aligned}$$

By analogue arguments as in Example 1, X is a measurable function and thus a random variable and the pushforward measure reads $p_X(\{1\}) = p_X(\{0\})$.

In Example 2 we could also have used the σ -subalgebra $\mathcal{F}' = \{\emptyset, \Omega, \{1, 3, 5\}, \{2, 4, 6\}\}$, with the same random variable (because $f^{-1}(1) = \{2, 4, 6\}$, $f^{-1}(0) = \{1, 3, 5\}$, both contained in $\mathcal{F}'$).

Remark: random variable (measurable functions) act on $X : \Omega \rightarrow \mathbb{R}$ while the probability distributions ((pushforward) measures) are defined on the corresponding σ -algebras $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ and thus on sets. However, there is another notational perversion: $p_X(1)$ instead of $p_X(\{1\})$ are common for measures. Note that the pushforward measures are the same in the above example, while the random processes are very different (coin toss vs die). This is quite powerful and motivates constructing general probability distributions on $\mathcal{B}(\mathbb{R})$, i.e. the Binomial distribution in the above Example. Therefore in practice, the probability space $(\Omega, \mathcal{F}, p)$ is hardly specified and one talks of 'a random variables which is Binomial distributed'. Apparently random variables don't make sense without specifying the underlying probability space. So, this is a rather cryptic wording for 1) there is a underlying probability space, 2) there is a random variable (measurable function) such that it's pushforward measure is the binomial distribution (on $\mathcal{B}(\mathbb{R})$).

- **Indicator function.** Given a set Ω and $A \subseteq \Omega$, the indicator function $\mathbb{1}_A$ given by

$$\begin{aligned}\mathbb{1}_A : \Omega &\rightarrow \{0, 1\} \\ \omega &\mapsto \begin{cases} 1, & \text{if } \omega \in A \\ 0, & \text{otherwise} \end{cases}\end{aligned}$$

The indicator function is a random variable on a measurable space $(\Omega, \mathcal{F})$
 $\Leftrightarrow A \in \mathcal{F}$.

- For a **discrete** probability space $(\Omega, \mathcal{F}, P)$ the σ -algebra is usually the powerset $\mathcal{F} = 2^\Omega$. It is common to use capital P for the probability measure. It is enough to specify P on singletons $(\{\{x\} | x \in \Omega\})$ as the rest follows from the definition of a probability measure. Brackets for singleton measures are often sloppily omitted, $P(x)$ instead of $P(\{x\})$.

2 Stochastic process

Definition 1 (Stochastic Process). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(S, \mathcal{S})$ a measurable space. A stochastic process $X = (X_t)_{t \in T}$ is collection of random variables (elements)*

$$\{X_t : \Omega \rightarrow S\}_{t \in T},$$

indexed by a set T (typically $\mathbb{N}_0$, or $\mathbb{R}_+$). S is called state space.

Thus, a stochastic process is a map $X : \Omega, T \rightarrow S$, $(\omega, t) \mapsto X_t(\omega)$ such that $X_t(\omega)$ is a $\mathcal{F}$ - $\mathcal{S}$ measurable function. For a fixed ω the map $X(\cdot, \omega) : T \rightarrow S$, $t \mapsto X_t(\omega)$ is called **path** of a stochastic process. A stochastic process is called **stationary in the strict sense** if for any $t_1, \dots, t_m \in T$ and translation $t_{1+s}, \dots, t_{m+s} \in T$ the random variables $X_1, \dots, X_m$ and $X_{1+s}, \dots, X_{m+s}$ have the same probability distributions.

Definition 2 (Probability Kernel [3]). *Let $(\mathcal{S}_1, \bar{\mathcal{S}}_1)$ and $(\mathcal{S}_2, \bar{\mathcal{S}}_2)$ be measurable spaces. A probability kernel (or markov kernel) k from $(\mathcal{S}_1, \bar{\mathcal{S}}_1)$ to $(\mathcal{S}_2, \bar{\mathcal{S}}_2)$ is a map $k : \mathcal{S}_1 \times \bar{\mathcal{S}}_2 \rightarrow [0, 1]$ such that*

1. *$k(x, \cdot)$ is a probability measure on $(\mathcal{S}_2, \bar{\mathcal{S}}_2)$ for every fixed $x \in \mathcal{S}_1$*
2. *$k(\cdot, A)$ is $\bar{\mathcal{S}}_1$ - $\mathcal{B}([0, 1])$ -measurable for every fixed $A \in \bar{\mathcal{S}}_2$*

Notationally, k is also called $(\mathcal{S}_1, \bar{\mathcal{S}}_1) \rightarrow (\mathcal{S}_2, \bar{\mathcal{S}}_2)$ probability kernel (while operating only on $\mathcal{S}_1 \times \bar{\mathcal{S}}_2$). First, clarify the notation:

$$\begin{aligned} k(x, \cdot) : \bar{\mathcal{S}}_2 &\rightarrow [0, 1], \quad A \mapsto K(x, A) \\ k(\cdot, A) : \mathcal{S}_1 &\rightarrow [0, 1], \quad x \mapsto K(x, A). \end{aligned}$$

The idea behind the first requirement is to describe stochastic transitions; having arrived at $x_1 \in \mathcal{S}_1$ the next state $x_2 \in \mathcal{S}_2$ is sampled from $\sim k(x_1, \cdot)$. The second requirement essentially assures integrability, which is desirable to 1) construct products or compositions of probability kernels and 2) to extend the theory in case $(\mathcal{S}_1, \bar{\mathcal{S}}_1)$ is a probability space, i.e it is additionally equipped with probability measure (in other words: if we wish to make statements about transitions if the state x_1 is uncertain).

If $(\mathcal{S}_1, \bar{\mathcal{S}}_1, p)$ is a measure (probability) space and k is a $(\mathcal{S}_1, \bar{\mathcal{S}}_1) \rightarrow (\mathcal{S}_2, \bar{\mathcal{S}}_2)$ probability kernel then $p \otimes k$ is a (probability) measure on $(\mathcal{S}_1 \times \mathcal{S}_2, \bar{\mathcal{S}}_1 \otimes \bar{\mathcal{S}}_2)$ defined by

$$p \otimes k(A) = \int_{\mathcal{S}_1} \int_{\mathcal{S}_2} \mathbb{1}_A((x_1, x_2)) k(x_1, dx_2) dp(x_1), \quad (1)$$

where $A \in \bar{\mathcal{S}}_1 \otimes \bar{\mathcal{S}}_2$. So this allows to construct a joint probability measure from a probability kernel and an ('initial') probability measure p .

Turning to the **product kernel**, let k_1 is a $(\mathcal{S}_1, \bar{\mathcal{S}}_1) \rightarrow (\mathcal{S}_2, \bar{\mathcal{S}}_2)$ probability kernel and k_2 is a $(\mathcal{S}_2, \bar{\mathcal{S}}_2) \rightarrow (\mathcal{S}_3, \bar{\mathcal{S}}_3)$ probability kernel. Then $k_1 \otimes k_2$ is a probability kernel $(\mathcal{S}_1, \bar{\mathcal{S}}_1) \rightarrow (\mathcal{S}_1 \times \mathcal{S}_2, \bar{\mathcal{S}}_1 \otimes \bar{\mathcal{S}}_2)$ defined by

$$k_1 \otimes k_2(x, A) = \int_{\mathcal{S}_2} \int_{\mathcal{S}_3} \mathbb{1}_A((x_2, x_3)) k_2(x_2, dx_3) k_1(x_1, dx_2), \quad (2)$$

where $x \in \mathcal{S}_1 \times \mathcal{S}_2, A \in \bar{\mathcal{S}}_1 \otimes \bar{\mathcal{S}}_2$.

Theorem 1 (Ionescu-Tulcea). *tbd*

3 Markov decision process

The following definitions are based on [2]

Definition 3 (Markov decision model). *A Markov decision model is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, p)$, with*

1. *State space: $(\mathcal{S}, \bar{\mathcal{S}})$ is a measurable space with σ -algebra $\bar{\mathcal{S}}$*
2. *Action space: $(\mathcal{A}, \bar{\mathcal{A}})$ is a measurable space with σ -algebra $\bar{\mathcal{A}}$, which is constructed by requiring for every state $s \in \mathcal{S}$ a measurable space $(\mathcal{A}_s, \bar{\mathcal{A}}_s)$ called the action space of state s and declare*

$$\mathcal{A} = \bigcup_{s \in \mathcal{S}} \mathcal{A}_s$$

$$\bar{\mathcal{A}} = \sigma \left(\bigcup_{s \in \mathcal{S}} \bar{\mathcal{A}}_s \right)$$

3. *Reward: $(\mathcal{R}, \bar{\mathcal{R}})$ is a measurable space with σ -algebra $\bar{\mathcal{R}}$, where $\mathcal{R} \subseteq \mathbb{R}, 0 \in \mathcal{R}$ and it's restricted Borel- σ -algebra $\bar{\mathcal{R}}$.*
4. *Reward function: p is a $(\mathcal{S} \times \mathcal{A}, \bar{\mathcal{S}} \otimes \bar{\mathcal{A}}) \rightarrow (\mathcal{S} \times \mathcal{R}, \bar{\mathcal{S}} \otimes \bar{\mathcal{R}})$ probability kernel*

$$p : \bar{\mathcal{S}} \otimes \bar{\mathcal{R}} \times (\mathcal{S} \times \mathcal{A}) \rightarrow [0, 1],$$

$$(B, (s, a)) \mapsto p(B; s, a)$$

Todo: this definition of a probability kernel is just the way around as above (definition 2) – fix this!

The key ingredient of the definition is the kernel p with the following interpretation. If the system is currently in state s and action a is played, then $p(\cdot; s, a)$ is the joint distribution of the next state s' and the reward r obtained. If all sets are discrete then $p(s', r; s, a)$ is the probability to obtain reward r and go to state s' if action a was taken in state s . According to the definition the reward can depend on the current state/action pair and the next state, but in most examples the reward and the next state will be independent.

A Markov decision model does not fully describe a process that runs from state to state and returns rewards. Only the transitions $(s, a) \mapsto (s', r)$ is described but not how the next action a' is chosen. Therefore and additional ingredient needed, the policy that can be chosen by the actor

Definition 4 (Policy). *For a Markov decision model $(\mathcal{S}, \mathcal{A}, \mathcal{R}, p)$ a policy is*

1. *An initial Markov kernel π_0 on $\bar{\mathcal{A}} \times \mathcal{S}$*
2. *A sequence of probability kernels $\pi = (\pi_t)_{t \in \mathbb{N}}$ on $\bar{\mathcal{A}} \times ((\mathcal{S} \times \mathcal{A})^{t-1} \times \mathcal{S})$ such that*

$$\pi_t(\mathcal{A}_s; s_0, a_0, \dots, s_{t-1}, a_{t-1}, s) = 1$$

for all $(s_0, a_0, \dots, s_{t-1}, a_{t-1}, s) \in (\mathcal{S} \times \mathcal{A})^{t-1} \times \mathcal{S}$ The set of all policies is denoted by Π .

Definition 5 (Markov decision process).

References

- [1] Bright Side of Maths. Bright Side of Maths youtube channel, 2025.
- [2] Leif Döring. The mathematics of reinforcement learning. Lecture Notes, 2024. Course: MA 544.
- [3] Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.
- [4] Terence Tao. *An Introduction to Measure Theory*, volume 126 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, Rhode Island, 2011.